| Question | Answer | Marks |
|---|---|---|
| 8(a) | photon energy (to remove electron) | **B1** |
| | minimum energy to remove electron<br>**or**<br>energy to remove electron from surface<br>**or**<br>energy to remove electron with zero kinetic energy | **B1** |
| 8(b)(i) | photon energy = $hf$ | **C1** |
| | number per unit time = $8.36 \times 10^{-3} / (1.36 \times 10^{15} \times 6.63 \times 10^{-34})$<br>$= 9.27 \times 10^{15}\ \text{s}^{-1}$ | **A1** |
| 8(b)(ii) | $hf = \Phi + E_{\text{MAX}}$ | **C1** |
| | $\Phi = (1.36 \times 10^{15} \times 6.63 \times 10^{-34}) - (3.09 \times 10^{-19})$<br>$= 5.93 \times 10^{-19}\ \text{J}$ | **A1** |
| 8(c)(i) | greater photon energy (and same work function) | **M1** |
| | so maximum kinetic energy is increased | **A1** |
| 8(c)(ii) | (greater photon energy and same power so) lower number of photons (per unit time) | **M1** |
| | (each electron absorbs one photon) so lower rate of emission | **A1** |

| Question | Answer | Marks |
|---|---|---|
| 9(a) | total power of radiation emitted (by the star) | B1 |